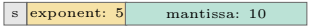
## (a) fp16, a power-of-two scale per element



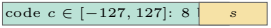

The scale puts $\sigma$ just below the top of the half-precision range. Every binade holds $2^{10}$ values, so each number is rounded relative to itself, $2^{-11}$, small ones included.




## (b) int8, one scale per element

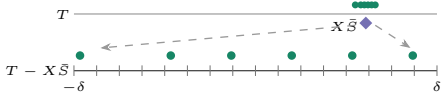



255 levels, $s = \sigma/127$ apart (every 16th drawn): the absolute rounding $s/2$ of every number is set by the element's largest entry; the small value rounds to zero.




per element

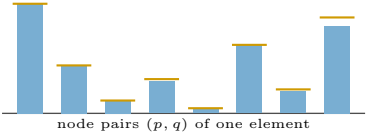
## (c) int8 mean and deviation



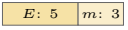

The element mean $\bar{S}$ in fp32, one number per component; the deviations coded $\delta/127$ apart, $\delta$ the variation of the tangent over the element.




## (d) a factor per node pair




$K_{pq} \approx s_e\, r_{pq}\, C_{pq}$: each pair spends the code range on its own block.




$r_{pq} = 2^{-E}(1 + m/8)$, then 9 codes $C_{pq}$